

Videos2Doc: Generating Documents from a Collection of Procedural Videos

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ABSTRACT

The availability of user-generated multi-modal content, including videos and images, in abundance makes it easy for users to use it as a reference and source of information. However, several hours may be required for the consumption of this large corpus of data. Particularly, for authors and content creators abstracting out information from videos to then representing it in a textual format is a tedious task. The challenges are multiplied due to the diversity and the variety introduced when there are several videos associated with a given query or topic of interest. We present, Videos2Doc, a machine learning-based framework for automated document generation from a collection of procedural videos. Videos2Doc enables author-guided document generation for those looking for authoring assistance and an easy consumption experience for those preferring the text or document media over videos. Our proposed interface allows the users to choose several visual and semantic preferences for the output document allowing the generation of custom documents and webpage templates from a given set of inputs. Empirical and qualitative evaluations establish the utility of Videos2Doc as well as the superiority over the current benchmarks. We believe, Videos2Doc will ease the task of making multimedia accessible through automation in conversion to alternate presentation modes.

CCS CONCEPTS

• **Computing methodologies** → **Natural language generation; Image representations; Video segmentation**; • **Information systems** → **Multimedia and multimodal retrieval; Video search**.

KEYWORDS

applications of intelligent user interfaces, assistive technologies, information retrieval, search, recommendation system, and modeling approaches

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1 INTRODUCTION

Multi-modal content such as videos and images are an important source of information that is available in abundance. Although availability of multiple videos for a given topic of interest brings in diversity in terms of different perspectives or methods of employing the same thing, it takes several hours for consuming such a rich content, there may be situations such as cataloging or for accessibility reasons that a document-like version of the content is convenient and preferred. Digesting several hours of content is tedious for abstracting out information from videos to then representing it in a textual format. For e.g. a blog writer, writing a food recipe blog on ‘Kung Pao Chicken’, will typically read and watch various resources available to gather information on the process: important steps, required ingredients, and any other relevant tips. Then, she will come up with an initial version which might require several iteration of edits before publishing the final blog. The variety of preferences that a user may have introduces several

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(a) Sample output for 'Kung Pao Chicken', inputs are recipe videos²

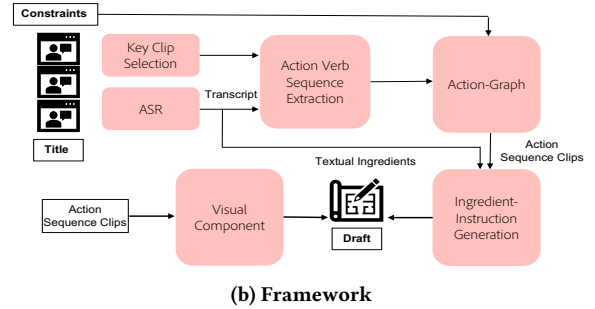


Figure 1: Videos2Doc

possibilities towards representing the same content - making it even more manually taxing. Here, we introduce an interface that takes in a set of procedural videos, a title, an output format and content preferences as input, and automatically generates a document-like formatted text-image version for the user's easy consumption. Our framework, Videos2Doc, also allows the user to select preferences and edit the output allowing customization of the output¹. Figure 1a shows a sample output and 1b illustrates the workflow diagram of Videos2Doc. Existing works include approaches to convert a video or an image(s) to text for procedural steps [9, 10, 13, 16–18] including the food recipe and the how-to instructions domain. However, limited work exists for converting multiple videos into a document-like media. Our approach addresses this while providing controls and custom knobs for the output space. Our examples and experiments use the food recipe domain for illustration. Videos2Doc enables author guided document generation for assisting authors with content creation and easy consumption experience for users who prefer text or document media over videos.

Prior Art: H. Lee et al. [5] and Bieñ et al. [1] focus on generating instructions using only ingredients or ingredients and recipe title by fine-tuning large language models like GPT2 [12]. Marin et al. [9] introduce the Recipe1M+ dataset and present an image-recipe retrieval task. Salvador et al. [13] generate ingredients and instructions for a recipe from an image of a final dish using an encoder-decoder-based model. Nishimura et al. [10] learn a shared embedding space for image and text. However, these works do not take video as an input. There exist works which given a single video extract recipes via object recognition and text generation [16], generate instructions as a dense-video captioning task [18], or extract a series of tuples (e.g. (heat, cast iron skillet), (fry, bacon, with heated skillet)) representing a procedure, and a list of key video clips with corresponding sentences to which the triplets are grounded, using a masked transformer [17]. None of these allow multiple videos as input or allow users to control the output. Videos2Doc also includes

common concepts in the video analysis literature: Key Frame Extraction and Action Segmentation. We re-implement the neural key clip selection model [17] and use an existing model [15] to extract action verbs in the transcript of a video.

2 VIDEOS2DOC: GENERATING DOCUMENT FROM VIDEOS

We propose a pipeline-based machine learning framework (Figure 1b) that takes in a collection of multi-modal user-generated content (videos, document title, and user-specified format) as input to generate a recipe document.

We first **extract transcripts and key clips** from recipe videos where an action is performed, using a neural key clip selection [17] which is a binary classifier that computes a combined representation for transcript (using BERT [2]) and video clip (using ResNet-50 [7]). This is followed by **extracting verbs** from action clips using a BERT-based semantic-role labeling (SRL) [4] model already fine-tuned on cooking domain. These verbs are then used to **create an action graph** representing verbs as nodes and probability of transition from one verb to another as edge weights. The best action verb sequence path is then identified by using a combination of optimized Depth-first search ($O(V + E)$) and Dijkstra [3] ($O(E + V \log V)$) algorithm and mapped back to video clips to obtain the corresponding **best frames**. The selected frames are clustered within an action group and the centroid of resulting cluster is chosen as the representative frame. Existing state of the art transcript generator is used to convert speech from videos to text³.

An encoder-decoder-decoder-based module [13] **generates ingredients** for each frame using an ingredient decoder. The generated ingredients along with *textual ingredients* (extracted from the transcripts and ingredients present in FoodKG [6], Recipe1M+ [9], and Tasty [14]) for each of the frames, and the frames are fed to an instruction decoder to **generate the instructions** corresponding to each frame denoting a step (instruction) of a recipe. Note that the *textual ingredients* are added to account for any missed ingredients

¹Demo video submitted along with this paper.

²<https://www.youtube.com/watch?v=0hVAuvPAIzA>, <https://www.youtube.com/watch?v=ntiGX3X-spA>, <https://www.youtube.com/watch?v=okvTq8gtr8c>

³<https://azure.microsoft.com/en-in/services/cognitive-services/speech-to-text/>

Table 1: Evaluation of instruction & ingredient generation on Tasty. Inst-V (& Ingr-V) denotes the size of instruction (& ingredient) vocabulary. SeqLen denotes maximum instruction length per frame. P, R, and F1 scores for Model-sq15, Model-sq20, and Model-sq25 are same as they use the same ingredient decoder while different instruction decoders. Bold-face denote best in each block and * best overall. Model-sq25 outperforms Inverse cooking across all the metrics (except precision) at *recipe* level.

Models	Granularity	Inst-V	Ingr-V	SeqLen	P	R	F1	BLEU	ROUGE-1	ROUGE-2	Coh-1	Coh-2
Inverse Cooking [13]	Recipe	23231	1488	20	55.95*	20.12	29.61	23.09	14.96	2.29	53.53	68.52
Model-sq15	Frame	24614	1861	15	36.26	23.70	28.67	16.33	35.59	10.86	77.38	78.61*
Model-sq20		24614	1861	20	36.26	23.70	28.67	16.48	35.41	10.98	84.90	78.45
Model-sq25		24614	1861	25	36.26	23.70	28.67	16.72	36.02	11.10	84.88	77.92
Model-sq15	Recipe	24614	1861	15	42.29	55.24	47.91	25.03	37.60	10.71	75.17	72.59
Model-sq20		24614	1861	20	42.29	55.24	47.91	25.71	39.69	11.07	83.85	72.41
Model-sq25		24614	1861	25	42.29	55.24*	47.91*	26.72*	40.30*	11.36*	85.72*	75.15

by the ingredient decoder. This is an advantage of using multi-modal inputs where different modalities (text, image, and audio) can complement each other. The encoder-decoder-decoder model is originally trained to generate ingredients and instructions given a final dish image. However, we extend it to generate ingredients and instructions at an intermediate frame level (i.e. the image may not contain a completed dish image). This is post-processed to ensure coherence between the ingredients and instructions.

We train this module in two-stages. We partially initialize the encoders and decoders with the weights from the pre-trained model of Salvador et al. [13] for faster training. We prepend the instructions and ingredients with a START token and append an END token at the end of an instruction or a list of ingredients. Image encoder and ingredient decoder are trained by minimizing a combination of following loss functions: **Binary cross entropy loss** is computed between the predicted (after pooling) and ground-truth ingredient for an intermediate frame as $L_{bce} = \frac{1}{N} \sum_{i=1}^N -y_i \log(p_i) + (1 - y_i) \log(1 - p_i)$, where y_i is the ground-truth, p_i the probability of being a key clip from the model, and N the number of training samples; $L_{ingr} = -\sum_{i=1}^V (z_i \log(\hat{z}_i) + (1 - z_i) \log(1 - \hat{z}_i))$. Here, V is the size of the vocabulary of ingredients, z_i denotes if ingredient i is correct for a recipe, and $\hat{z}_i$ is prediction from the ingredient decoder; **Cardinality loss** penalizes the model if the cardinality of predicted ingredients is less or more than the ground truth cardinality. It is computed as l_1 penalty; **Eos loss** teaches the model to learn the stopping criteria of the ingredient prediction. As inclusion of EOS (end of sentence) token in the pooling operation would result in losing the information of where the token appears, eos loss is used. It is calculated as the binary cross-entropy loss between the predicted eos probability at all time-steps and the ground truth. In the second stage, we jointly train the ingredient encoder and instruction decoder by minimizing the negative log-likelihood and updating θ_R and θ_E , where θ_R and θ_E are the parameters of ingredient encoder and instruction decoder, respectively.

A **visual component selection** module retrieves images corresponding to a recipe to complete the final document. Our pipeline is capable of generating a **text heavy** document by retrieving a hero image as well as a **visual heavy** document by presenting the frames corresponding to each of the action verbs in the action verb sequence in the form of recipe cards, slider or a short video. Graph

traversal logic is implemented to allow user specified changes in the output.

3 EXPERIMENTS

Evaluation Metrics: We independently evaluate the neural key clip selection, and ingredient & instruction generation modules⁴. The ingredient & instruction generation is evaluated on the Tasty dataset [14] with gold-annotations obtained per frame (intermediate step). We report performance at both *frame* and *recipe* level. Average scores across all frames in a recipe are reported as *Frame* level scores, while *recipe* level scores are computed by first averaging scores for all the frames in a recipe and then averaging across all recipes. We report **Precision (P)**, **Recall (R)**, **F1** for assessing ingredient prediction along with **BLEU** [11] and **ROUGE-n**⁵ [8] for the quality of instructions. We introduce **Coherence** between the predicted ingredients and instructions calculated as, **Coh-1** measures the coherence of the predicted instructions w.r.t. the ingredients. It is computed as the percentage of predicted ingredients (IG) present in predicted instructions (IS) & **Coh-2** estimates the percentage of ingredients in predicted instructions (IS) covered in predicted ingredients (IG).

Results: We compare ingredient & instruction generation trained using intermediate frames against the inverse cooking model [13] in Table 1. Model-sq25 outperforms Inverse cooking across all measures at *recipe* level except for precision. Although precision is lower, Model-sq25 achieves higher F1 which indicates a greater balance between precision and recall for predicting ingredients. Higher BLEU and ROUGE indicate that the generated instructions are of better quality and closer to the ground truth instructions. Higher coherence score establishes the importance of generating instructions using intermediate frames rather than just the final recipe image. This shows the benefit of using temporal information present in video over a single image and attending to an intermediate recipe image for generating ingredients and instructions corresponding to each step.

4 CONCLUSION

We present a novel framework to generate a document from procedural videos, specifically, tested on the food recipe domain. Our evaluation through automated metrics shows that (1) Videos2Doc

⁴Neural key clip selection has an accuracy of 76.75% with precision, recall, and F1 of 0.73, 0.56, and 0.62 respectively.

⁵We use bleu_score and rouge_score from torchtext library

outperforms the original encoder-decoder-decoder Salvador et al. [13] model when evaluated on Tasty dataset (see Table 1); (2) feeding a larger number of videos to the pipeline generates highly coherent and comparable results to the ground truth; (3) the generated documents are almost equally recallable, adaptable and coherent as compared to input videos; and (4) Videos2Doc generates highly coherent, relevant, adaptable and correct documents.

REFERENCES

- [1] Michał Bień, Michał Gilski, Martyna Maciejewska, Wojciech Taisner, Dawid Wisniewski, and Agnieszka Lawrynowicz. 2020. RecipeNLG: A Cooking Recipes Dataset for Semi-Structured Text Generation. In *Proceedings of the 13th International Conference on Natural Language Generation*. Association for Computational Linguistics, Dublin, Ireland, 22–28. <https://aclanthology.org/2020.inlg-1.4>
- [2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* abs/1810.04805 (2018).
- [3] Edsger W Dijkstra. 1959. A note on two problems in connexion with graphs. *Numerische mathematik* 1, 1 (1959), 269–271.
- [4] Daniel Gildea and Daniel Jurafsky. 2002. Automatic labeling of semantic roles. *Computational linguistics* 28, 3 (2002), 245–288.
- [5] Helena H. Lee, Ke Shu, Palakorn Achananuparp, Philips Kokoh Prasetyo, Yue Liu, Ee-Peng Lim, and Lav R Varshney. 2020. RecipeGPT: Generative pre-training based cooking recipe generation and evaluation system. In *Companion Proceedings of the Web Conference 2020*. Association for Computing Machinery, 181–184.
- [6] Steven Haussmann, Oshani Seneviratne, Yu Chen, Yarden Ne’eman, James Codella, Ching-Hua Chen, Deborah L McGuinness, and Mohammed J Zaki. 2019. FoodKG: a semantics-driven knowledge graph for food recommendation. In *International Semantic Web Conference*. Springer, 146–162. https://doi.org/10.1007/978-3-030-30796-7_10
- [7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
- [8] Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*. 74–81.
- [9] Javier Marin, Aritro Biswas, Ferda Ofli, Nicholas Hynes, Amaia Salvador, Yusuf Aytar, Ingmar Weber, and Antonio Torralba. 2019. Recipe1m+: A dataset for learning cross-modal embeddings for cooking recipes and food images. *IEEE transactions on pattern analysis and machine intelligence* 43, 1 (2019), 187–203.
- [10] Taichi Nishimura, Atsushi Hashimoto, and Shinsuke Mori. 2019. Procedural text generation from a photo sequence. In *Proceedings of the 12th International Conference on Natural Language Generation*. 409–414.
- [11] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.
- [12] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. (2019).
- [13] Amaia Salvador, Michal Drozdal, Xavier Giro-i Nieto, and Adriana Romero. 2019. Inverse cooking: Recipe generation from food images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10453–10462.
- [14] Fadime Sener and Angela Yao. 2019. Zero-shot anticipation for instructional activities. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 862–871.
- [15] Peng Shi and Jimmy Lin. 2019. Simple bert models for relation extraction and semantic role labeling. *arXiv preprint arXiv:1904.05255* (2019).
- [16] Atsushi Ushiku, Hayato Hashimoto, Atsushi Hashimoto, and Shinsuke Mori. 2017. Procedural text generation from an execution video. In *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 326–335.
- [17] Frank F Xu, Lei Ji, Botian Shi, Junyi Du, Graham Neubig, Yonatan Bisk, and Nan Duan. 2020. A benchmark for structured procedural knowledge extraction from cooking videos. *arXiv preprint arXiv:2005.00706* (2020).
- [18] Luowei Zhou, Yingbo Zhou, Jason J Corso, Richard Socher, and Caiming Xiong. 2018. End-to-end dense video captioning with masked transformer. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 8739–8748.